

# **Digital design intern ADI**

## **FPGA assignment 2 co\_design NAND gate**

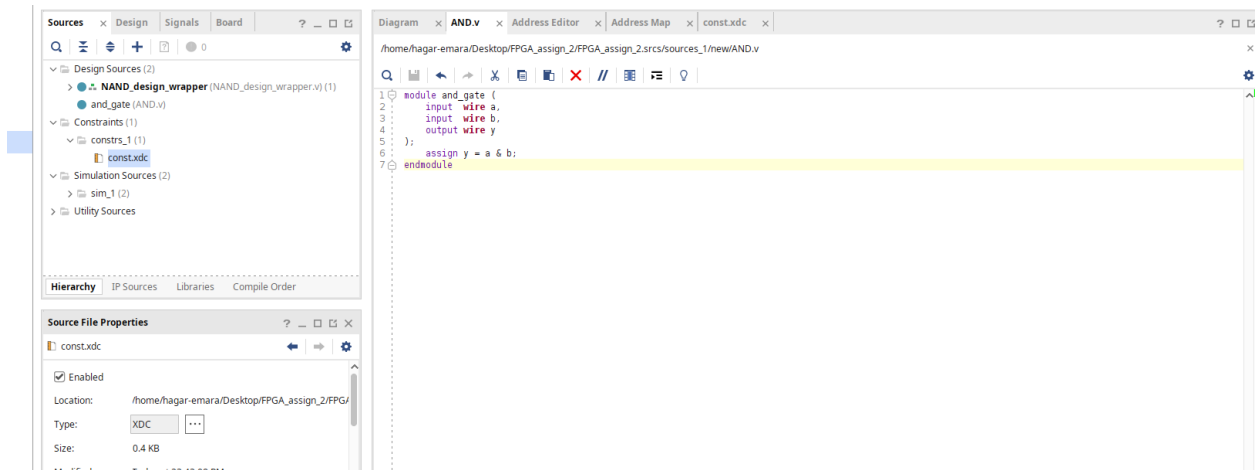
Made by:

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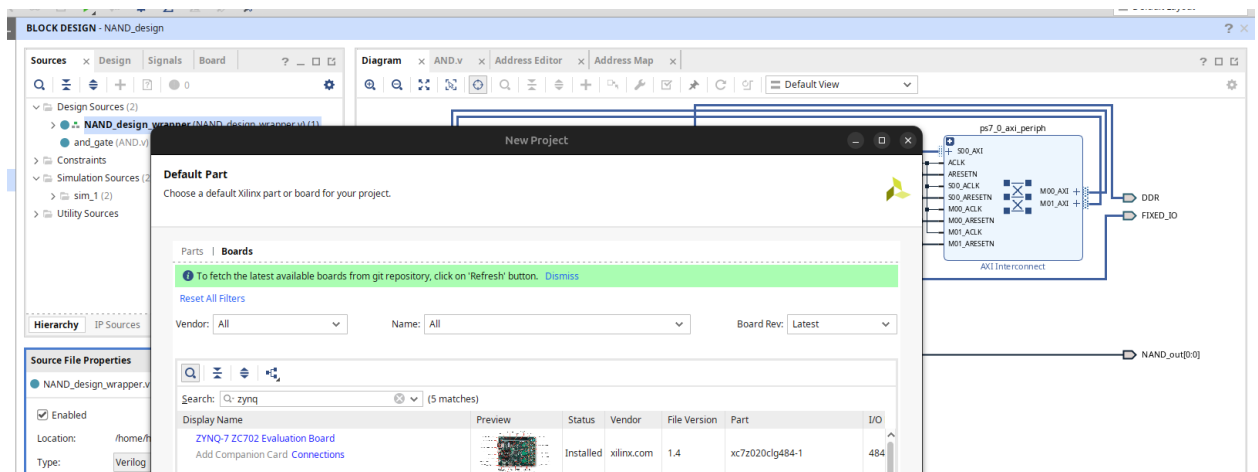
Submitted to:

Eng/ Fatma Ali

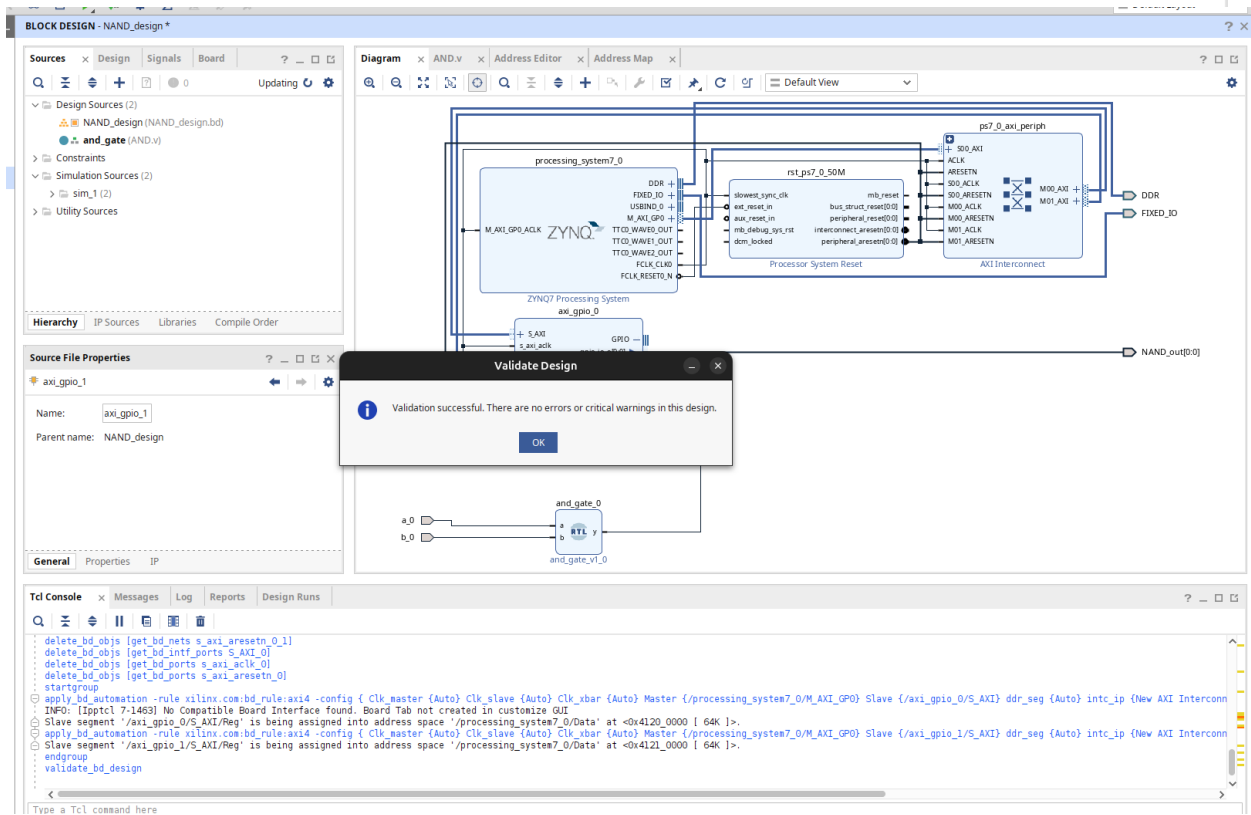
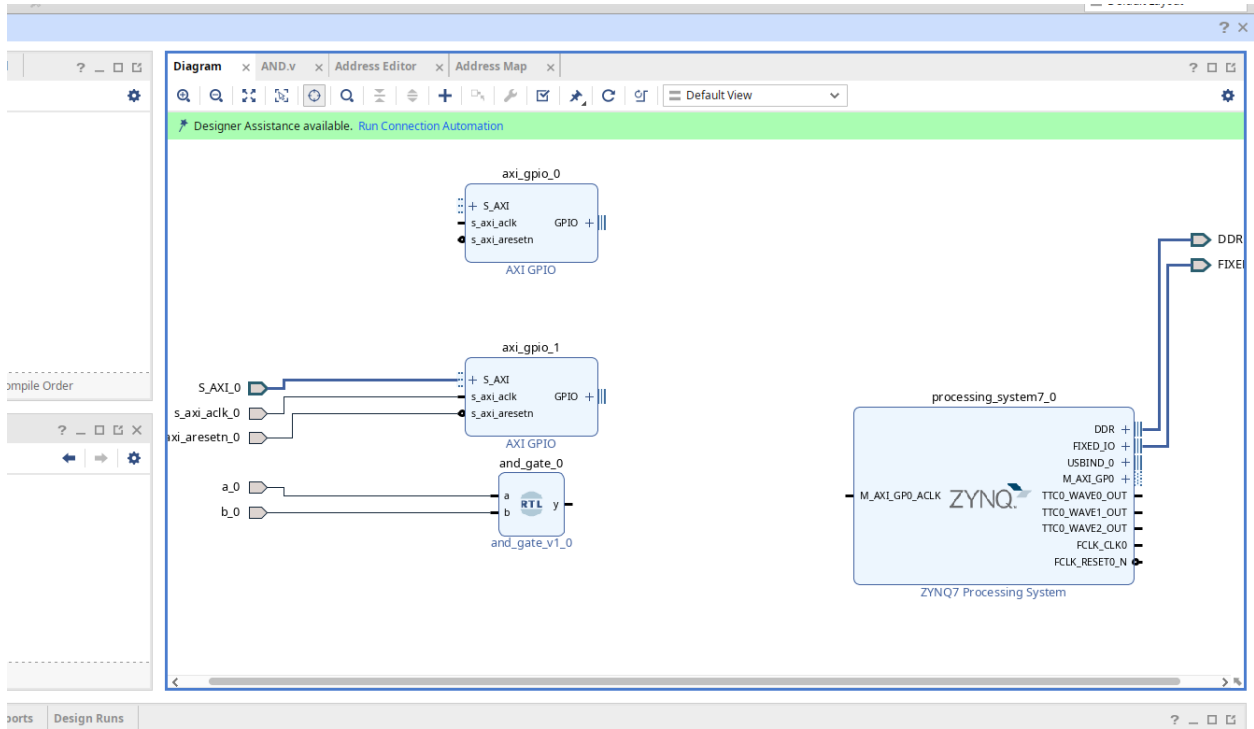
## AND code



## Chosen board



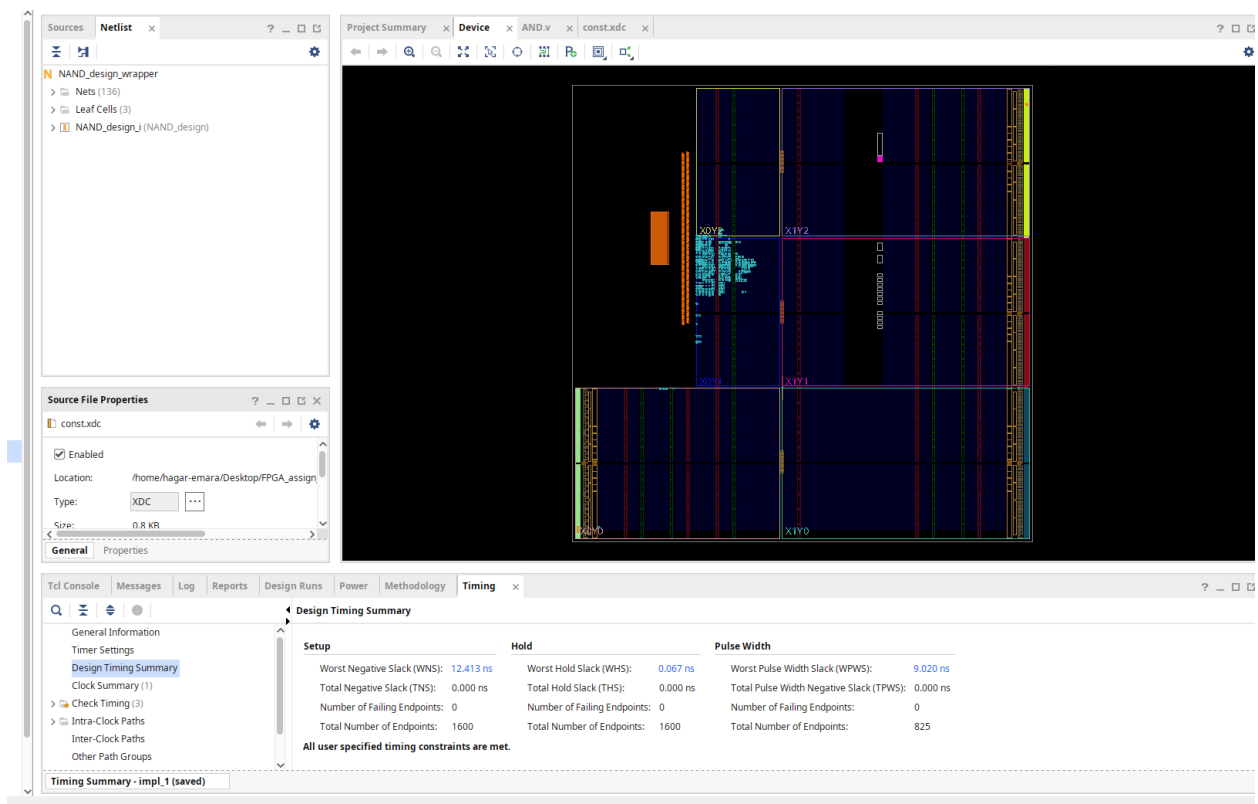
## Block diagram



## Constraints

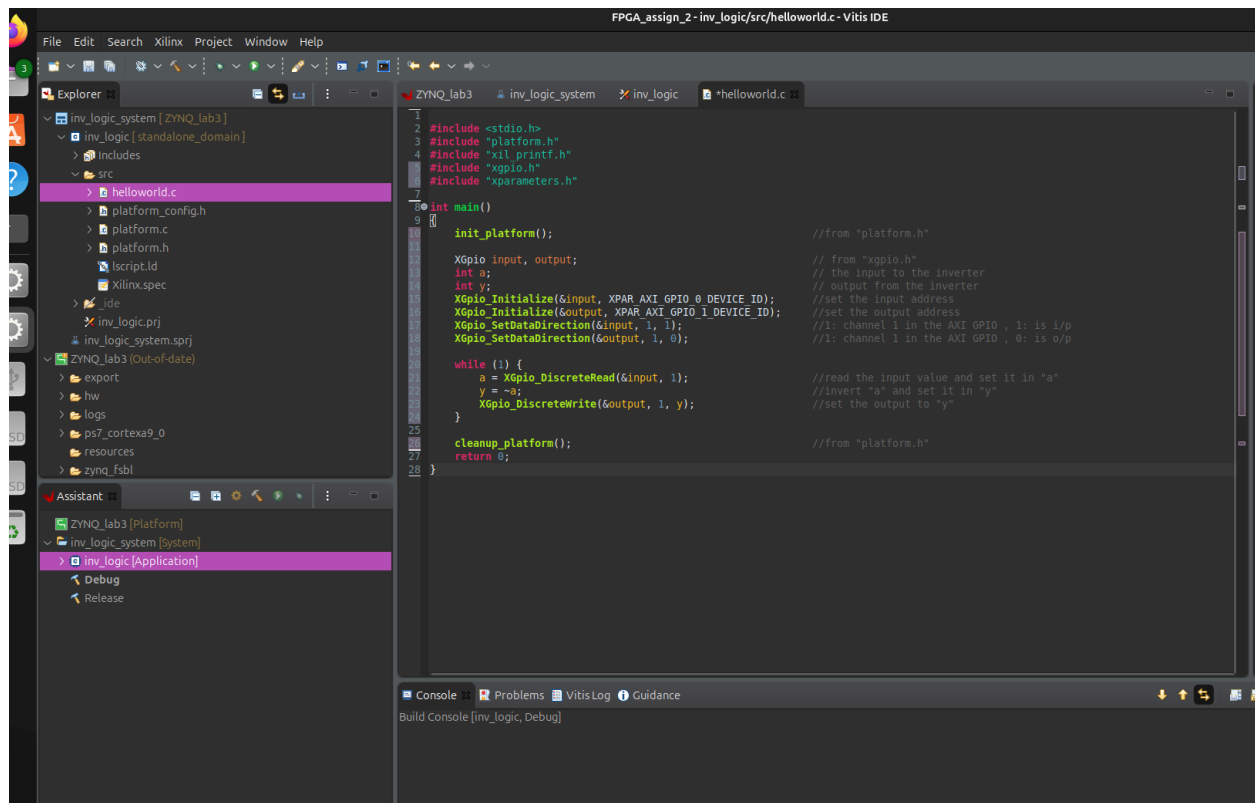
```
# -----  
# User DIP Switches - feeding AND gate inputs (PL fabric)  
# -----  
set_property PACKAGE_PIN W6 [get_ports {a}]; # "GPIO_DIP_SW0"  
set_property PACKAGE_PIN W7 [get_ports {b}]; # "GPIO_DIP_SW1"  
set_property IOSTANDARD LVCMOS25 [get_ports {a}];  
set_property IOSTANDARD LVCMOS25 [get_ports {b}];  
  
# -----  
# NAND result output - ZC702 has no PL-fabric LED, using unused PMOD pin instead  
# -----  
set_property PACKAGE_PIN E15 [get_ports {NAND_out[0]}]; # "PMOD1_0_LS"  
set_property IOSTANDARD LVCMOS25 [get_ports {NAND_out[0]}];
```

## After implementation



**Then, I exported the bitstream and opened a platform in vitis and then made a test application for the inverter as software code**

## Vitis



## build done successfully

